English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the original handwritten solutions. Apparent source inconsistencies are identified in notes.

Problem 1

Filling the first square takes Ge(1) trials; filling a second distinct square after the first takes Ge(8/9), and so on. Hence

$$\mathbb{E}[\text{time}] = \mathbb{E}[\text{Ge}(1) + \cdots + \text{Ge}(1/9)] = 9\left(1 + \frac{1}{2} + \cdots + \frac{1}{9}\right) \approx 25.46.$$

Let T_2 be the first time at which one square contains two darts. Split according to the square where this first repetition occurs. By symmetry,

$$\begin{aligned}\mathbb{E}T_2 &= 9\mathbb{E}[T_2 \mathbf{1}_{\{\text{first repetition occurs in a specified square}\}}] \\ &= 9\left[2\left(\frac{1}{9}\right)^2 + 3\binom{8}{1} 2! \left(\frac{1}{9}\right)^3 + 4\binom{8}{2} 3! \left(\frac{1}{9}\right)^4 + \cdots\right].\end{aligned}$$

In compact form,

$$\mathbb{E}T_2 = 9\sum_{k=0}^{8}(k+2)! \left(\frac{1}{9}\right)^{k+2} \binom{8}{k} \approx 4.46.$$

For an $n \times n$ grid, the answer to the first question is

$$n^2\left(1 + \cdots + \frac{1}{n^2}\right) \sim n^2 \ln(n^2) = 2n^2 \ln n.$$

For the second question, the handwritten source gives

$$n^2 \sum_{k=0}^{n-1}(k+2)! \left(\frac{1}{n^2}\right)^{k+2} \binom{n-1}{k}. \qquad \text{(Note 1)}$$

Problem 2

1. Since $(X_1, \ldots, X_n)$ is a permutation,

$$S = X_1 + \cdots + X_n = 1 + \cdots + n = \frac{n(n+1)}{2}.$$

Therefore $DS = 0$ and $\mathbb{E}S = n(n+1)/2$.

2.

$$\begin{aligned}D(X+Y) &= \mathbb{E}[(X+Y)^2] - (\mathbb{E}X + \mathbb{E}Y)^2 \\ &= \mathbb{E}X^2 - (\mathbb{E}X)^2 + \mathbb{E}Y^2 - (\mathbb{E}Y)^2 + 2(\mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y).\end{aligned}$$

3. By symmetry,

$$\mathbb{E}S = \mathbb{E}(X_1 + \cdots + X_n) = n\mathbb{E}X_1, \qquad \mathbb{E}X_i = \frac{n+1}{2}.$$

Similarly,

$$\mathbb{E}(X_1^2 + \cdots + X_n^2) = \mathbb{E}(1^2 + \cdots + n^2) = \frac{n(n+1)(2n+1)}{6} = n\mathbb{E}X_1^2,$$

so

$$\mathbb{E}X_i^2 = \frac{(n+1)(2n+1)}{6},$$

$$DX_i = \mathbb{E}X_i^2 - (\mathbb{E}X_i)^2 = \frac{n+1}{2}\left(\frac{2n+1}{3} - \frac{n+1}{2}\right) = \frac{n^2-1}{12}.$$

4.

$$0 = DS = nDX_1 + \binom{n}{2} 2\operatorname{Cov}(X_1, X_2)$$
$$= n\frac{n^2-1}{12} + n(n-1)\operatorname{Cov}(X_1, X_2).$$

Hence

$$\operatorname{Cov}(X_i, X_j) = \begin{cases} -(n+1)/12, & i \neq j, \\ (n^2-1)/12, & i = j. \end{cases}$$

Problem 3

1. Independence gives $D(M_1 + \cdots + M_n) = nDM_1$. Let $M = M_1 = \max(U_1, V_1)$. For $t \in [0,1]$,

$$\mathbb{P}(M \leq t) = \mathbb{P}(U_1 \leq t)\mathbb{P}(V_1 \leq t) = t^2, \qquad f_M(t) = 2t\mathbf{1}_{[0,1]}(t).$$

Therefore

$$\mathbb{E}M = \int_0^1 2t \cdot t \,\mathrm{d}t = \frac{2}{3}, \qquad \mathbb{E}M^2 = \int_0^1 t^2 \cdot 2t \,\mathrm{d}t = \frac{1}{2},$$
$$DM = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}, \qquad D(M_1 + \cdots + M_n) = \frac{n}{18}.$$

2. The source writes the CLT approximation as

$$M_1 + \cdots + M_n \approx N\left(n \cdot \frac{2}{3}, \frac{1}{18}\right). \qquad \text{(Note 2)}$$

Problem 4

1. The density normalization requires

$$\int_1^\infty \frac{c}{x^4}\,\mathrm{d}x = 1, \qquad \left[-\frac{c}{3}x^{-3}\right]_1^\infty = \frac{c}{3}, \qquad c = 3.$$

Next,

$$\mathbb{E}X_1 = \int_1^\infty \frac{c}{x^3}\,\mathrm{d}x = \left[-\frac{c}{2}x^{-2}\right]_1^\infty = \frac{c}{2} = \frac{3}{2},$$
$$\mathbb{E}X_1^2 = \int_1^\infty \frac{c}{x^2}\,\mathrm{d}x = c = 3, \qquad DX_1 = 3 - \frac{9}{4} = \frac{3}{4}.$$

Since $DX_1 < \infty$, the central limit theorem gives

$$\frac{S_n - n \cdot \frac{3}{2}}{\sqrt{3n/4}} \approx N(0,1), \qquad a = \frac{3n}{2}, \quad b = \frac{\sqrt{3n}}{2}.$$

2. Let $S = \sum_{i=1}^{40} X_i$. Here $\mathbb{E}S = DS = 400$. Using the CLT with $40 > 30$,

$$99\% \leq \mathbb{P}(S > a) = \mathbb{P}\left(\frac{S - 400}{20} > \frac{a - 400}{20}\right)$$
$$\approx \mathbb{P}\left(N(0,1) > \frac{a - 400}{20}\right).$$

From the normal table, $(a - 400)/20 \approx -2.33$, so

$$a = 400 - 46.6 = 353.4.$$

Transcription notes

1. For an $n \times n$ grid, the compact expression should use the $n^2 - 1$ other squares, so both occurrences of $n - 1$ in this source formula require review.

2. The preceding variance calculation gives $n/18$. The normal approximation in the original omits the factor n in its variance parameter.